### 7.3.4 Biological Process of Ageing

- The genes determine the lifespan, genes may have a role to play in the ageing process.
- Wear and Tear of important organs by continues functioning .
- Accumulation of toxic materials (E.g.. cholesterol) in the vital organ like heart, brain eye and thereby damaging them.
- Loss of important genetic material during DNA repair.
- Accumulate of stress over lifetime with its resultant effects.
- Exertion of production and deficiency of important hormones e.g. growth hormone and androgen, oestrogen and thyroid hormones.

Evaluation of ageing helps to explain why survival, reproductive success, and functioning of almost all living organism decline at old age.

## 7.4 EVOLUATIONARY BASIS OF AGEING

### 7.4.1 Ageing theories which mainly has been categorise into two main categories

#### Error Theory

Ageing is a result of internal or external assaults that damage cells or organs so they can no longer function properly.

#### Programmed Theory

Ageing has a biological time table or internal biological clock

#### Error Theory

a) Waste accumulation theory

b) Mitochondrial damage theory

#### a) Waste Accumulation Theory

The human lifespan simply reflects the level of free radical damage that accumulates in cells when enough damage accumulates, cells cannot survive properly any more and they just simply give up.

Genetic mutations occur and accumulate with increasing age, causing cells to deteriorate and malfunction RNA molecule is highly susceptible to oxidative stress

#### b) Mitochondrial Damage Theory

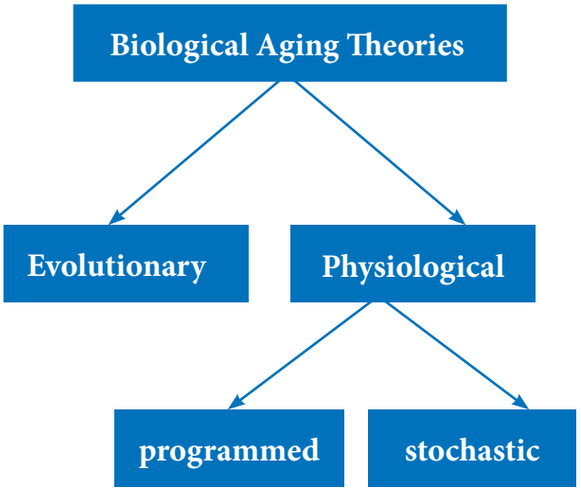
It is well known that mitochondrial DNA is much more oxidized with age than nuclear DNA reactive oxygen species are continuously generated in the mitochondrial electron transport chain.

#### 2. Programmed Theory

#### a. Immunological Theory

The immune system is programmed to decline over time which leads to an increased vulnerability to infections, disease and thus aging and death. It is well documented that the effectiveness of the immune system peaks at puberty and gradually declines there after with advance in age. eg. as one grows older antibodies lose their effectiveness.

### 7.4.2 Biological Aging Theories



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